COMPUTER VISION • GROUP 17

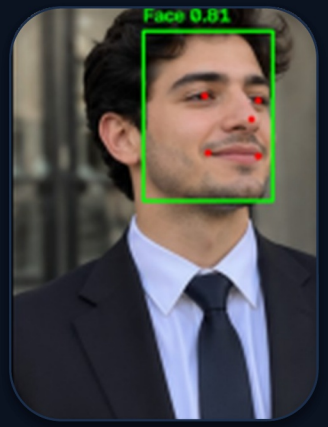
Face and Eye Detection

Haar Cascade vs. Deep Learning Models

A practical comparison of traditional feature cascades, SCRFD face detection, and a YOLO-based eye detector.

13 FACE IMAGES

10 EYE IMAGES



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Why detect faces and eyes?

Two localization tasks, two model families, one shared evaluation setup.

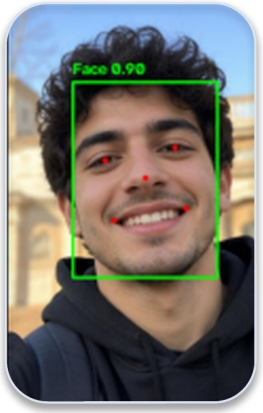
THE TASKS

Turn pixels into useful locations.

Face detection finds each face in an image. Eye detection focuses on smaller eye regions, where scale, pose and lighting can make localization more delicate.

PROJECT OBJECTIVES

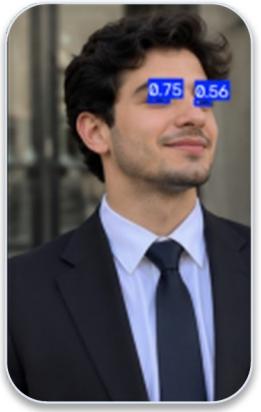
Understand • Compare • Identify trade-offs



FACE DETECTION

Haar Cascade vs. SCRFD

Whole-face bounding boxes; SCRFD can also return five facial landmarks.



EYE DETECTION

Haar Cascade vs. YOLO model

Compare detected eye-region boxes. Confidence is shown; labels are suppressed in the supplied plot.

Focus: observable outputs, not unsupported accuracy claims.

Dataset and experimental setup

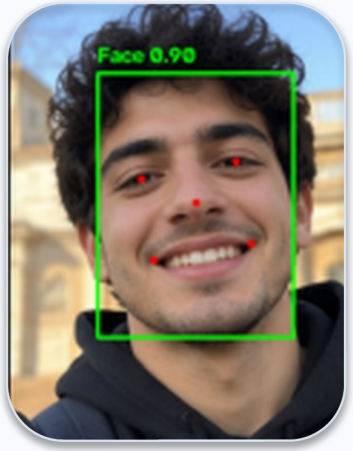
Same images per task make side-by-side visual comparisons more meaningful.

13 FACE IMAGES
10 initial + 3 crowd scenes

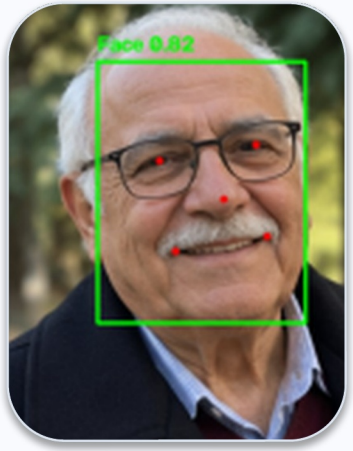
10 EYE IMAGES
Initial images only

2 METHODS / TASK
Same inputs per comparison

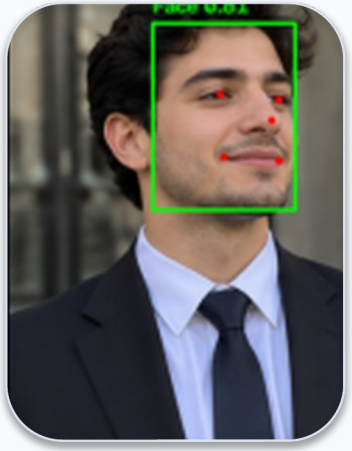
SAMPLE CONDITIONS



Single face



Older portrait



Portrait in profile

STRESS TEST



Crowded scene • Image 12

+2 CROWD SCENES

Images 11 and 13 are included in face detection only.

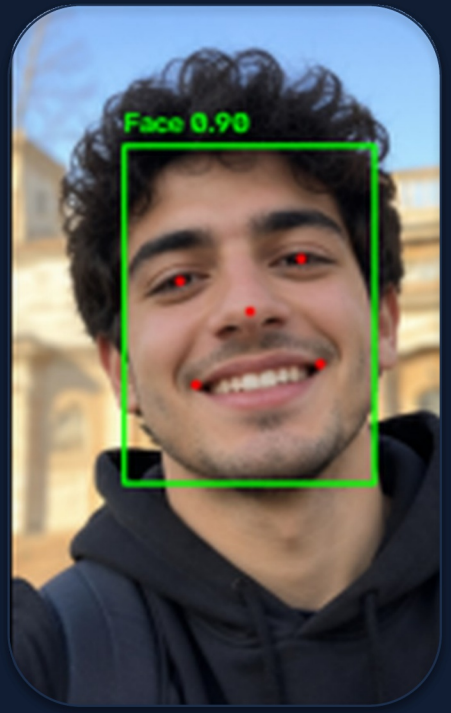
Eye evaluation uses images 1–10 only.

Small, illustrative set — not a statistically representative benchmark.

Face detection is not eye detection

Different target sizes require different visual decisions.

01 FACE DETECTION



Find the whole face

Output: a bounding box and a detection score. SCRFD also returns five facial landmark points in this implementation.

BOUNDING BOX

02 EYE LOCALIZATION



Find smaller regions

Output: eye-region boxes and confidence scores. The supplied plotted images hide class labels, so they do not visibly establish open/closed predictions.

CONFIDENCE ≠ CORRECTNESS

Haar Cascade: the traditional baseline

A sliding-window detector built from simple visual patterns and a sequence of tests.

01

Haar-like features

Compare light and dark rectangular regions.

02

Integral image

Reuse summed pixels to compute features quickly.

03

AdaBoost

Select and combine useful weak features.

04

Cascade stages

Reject unlikely windows early; keep candidates.

WHY IT IS STILL USEFUL

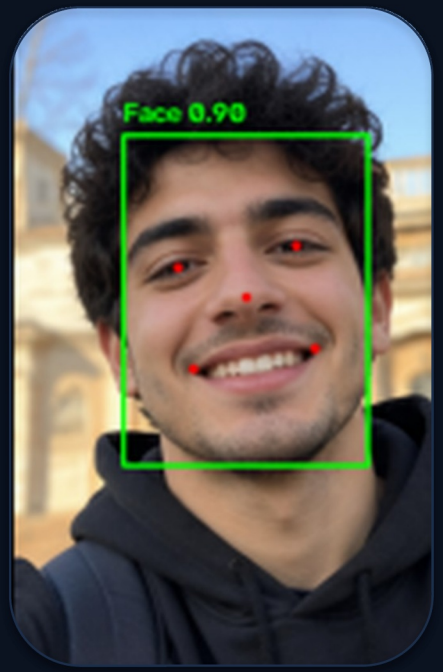
- Lightweight and simple to deploy
- Built into OpenCV; no neural model setup
- Can be a useful baseline for controlled scenes

WHERE IT CAN STRUGGLE

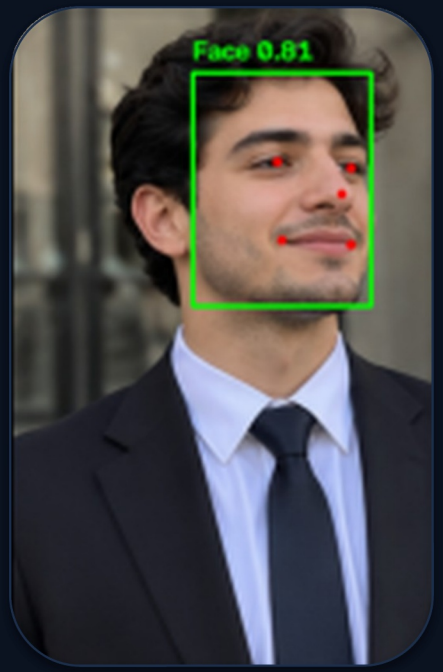
- Pose, lighting and partial occlusion
- Small or distant faces; busy backgrounds
- False positives — including boxes off the target

SCRFD: deep learning for face detection

The supplied code loads a pretrained ONNX model and runs it through InsightFace.



Single face



Profile view

MODEL DETAILS

SCRFD-face-detection-ONNX

10g_bnkps.onnx • downloaded via Hugging Face

- Deep model features support face localization across varied sizes and conditions.
- Returns bounding boxes, scores and five landmark points per face (when available).
- Configured for CPU inference: ctx_id = -1; input_size = 640 × 640; det_thresh = 0.5.

A stronger visual fit in selected challenging cases is an observation from this dataset—not a universal guarantee.

Implementation: SCRFD face detection

The code below follows the supplied Python implementation.

1 DOWNLOAD + INITIALIZE

```
weights_path = hf_hub_download(
    repo_id="RuteNL/SCRFD-face-detection-ONNX",
    filename="10g_bnkps.onnx"
)
detector = get_model(weights_path)
detector.prepare(
    ctx_id=-1, input_size=(640, 640), det_thresh=0.5
)
```

Downloads the pretrained ONNX weights and configures CPU inference.

2 RUN INFERENCE

```
bboxes, kpss = detector.detect(
    img, input_size=(640, 640), max_num=0
)
```

WHAT THE OUTPUT CONTAINS

bbox	x1, y1, x2, y2 + score
landmarks	five points per face
count	len(bboxes)

Visualization OpenCV draws green rectangles and score text; red circles mark the returned facial landmarks.

Face results: the first 10 images

Observed box counts from the provided outputs; counts are not verified accuracy metrics.

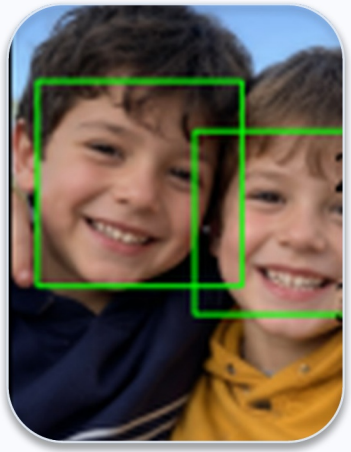
15 Haar boxes

17 SCRFD detections

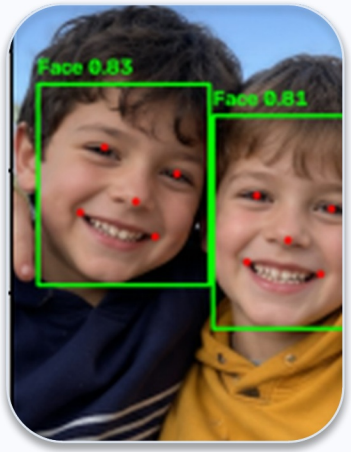
IMAGE	HAAR	SCRFD	COUNT BARS (MAX = 3)
01	1	1	
02	0	1	
03	1	3	
04	3	3	
05	1	1	
06	3	2	
07	1	1	
08	1	1	
09	3	3	
10	1	1	

SELECTED EXAMPLES

Image 6 • two faces visible



Haar Cascade



SCRFD

2 vs 3

SCRFD reports two faces; Haar produces an additional box on clothing.

Key observations

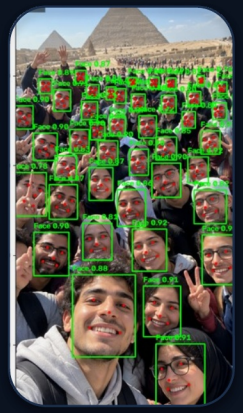
- Image 2: Haar missed a face found by SCRFD.
- Image 3: SCRFD found 3 vs. Haar's 1.
- Both totals are close to the visually assessed 17 faces—but equal totals would not prove accuracy.

Crowded scenes: a more demanding test

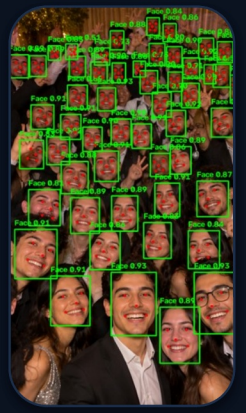
The detector output shows small and distant faces across dense scenes.



IMAGE 12 • STAIRCASE CROWD



11



13

Additional crowd outputs

SCRFD results were reviewed for all three scenes; the two supplemental scenes are kept small here.

REPORTED DETECTION COUNTS

44	72	52
logged count 1	logged count 2	logged count 3

TOTAL **168**

Counts are shown in logged order; the supplied snippet does not preserve a verifiable filename-to-count mapping. No exact Haar counts are claimed.

Eye model: localization with a YOLO interface

The supplied code loads model.pt and uses Ultralytics for prediction.



Male portrait examples from the supplied model output

MODEL + INFERENCE CONFIGURATION

MichalMlodawski/open-closed-eye-detection

Weights: model.pt • Loaded with YOLO(weights_path)

Confidence threshold	**0.25**
Inference image size	**960**
Plot output	**boxes + confidence; labels=False**
Important interpretation	The repository name mentions open/closed eyes and the code loads model.names, but the supplied plots hide class labels. The screenshots therefore demonstrate eye-region boxes and confidence values—not visibly verified open/closed classifications.

Implementation: eye detection

The excerpt is taken from the supplied model-loading and prediction code.

1 DOWNLOAD + LOAD

```
weights_path = hf_hub_download(
    repo_id="MichalMlodawski/open-closed-eye-detection",
    filename="model.pt"
)
model = YOLO(weights_path)
print("Classes:", model.names)
```

2 PREDICT + VISUALIZE

```
results = model.predict(
    source=path, conf=0.25, imgsz=960
)
result = results[0]
annotated_image = result.plot(
    labels=False, conf=True
)
```

PREDICTION FLOW

Input — Image path

YOLO — Run detector

Boxes — Read result.boxes

Plot — Draw boxes + scores

`labels=False` intentionally hides class names in the annotated result. Confidence scores remain visible in the provided screenshot.

Eye results: 10 images only

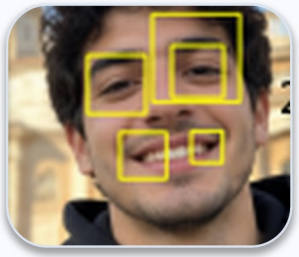
Approximate visible box counts transcribed from the supplied composite screenshots.

IMAGE	HAAR	YOLO MODEL	PAIR BARS
01	4	2	
02	4	2	
03	5	6	
04	8	6	
05	0	2	
06	5	4	
07	3	2	
08	1	2	
09	5	6	
10	2	2	

Totals: 37 Haar boxes vs. 34 model boxes — estimates only.

SELECTED VISUAL COMPARISONS

IMAGE 1



Haar



YOLO

IMAGE 5





IMAGE 10





Read carefully

More boxes do not automatically mean better detection. Inspect whether boxes align with eyes and whether eyes were missed.

No ground-truth annotations were supplied.

Evaluation: what the evidence supports

Visual findings are useful, but formal detection metrics require ground truth.

FACE DETECTION

Haar Cascade vs. SCRFD

- SCRFD found faces missed by Haar in selected initial images.
- On crowded scenes, SCRFD boxes span more small/distant faces in the screenshots.

EYE DETECTION

Haar Cascade vs. YOLO model

- Some Haar boxes are misaligned or appear on non-eye regions.
- The model shows confidence-tagged boxes, but label suppression limits visible class interpretation.

METRICS FOR A RIGOROUS COMPARISON

Precision

Correct predictions ÷ all predictions

Recall

Detected ground-truth objects ÷ all actual objects

F1-score

Harmonic mean of precision and recall

Inference time

Elapsed time per image under controlled conditions

Ground-truth boxes/labels are required to calculate reliable precision, recall and F1. No benchmark score or speed winner is claimed here.

Conclusion and next steps

Findings apply to this small experiment and its supplied screenshots.

WHAT THIS EXPERIMENT SHOWED

01	**Haar Cascade**	Lightweight and simple, but susceptible to missed faces and false positives in difficult scenes.
02	**SCRFD**	Produced more comprehensive-looking detections in several tested cases, especially crowds.
03	**Eye model**	Localized eye regions with confidence values; open/closed behavior is not visibly confirmed in the plots.
04	**Evaluation**	Verified ground truth and controlled timing are needed for defensible performance metrics.

SOURCE CODE AND IMPLEMENTATION

GitHub Repository

Open GitHub Repository

Future work

Label ground truth • test more images
measure inference time on the same hardware

Conclusion: deep learning approaches looked more effective in selected examples, not necessarily under every condition.

Thank You!

Questions?

Made by Ahmed Youssef

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